BlackOps: GIS-Informed Planning for Healthcare Resilience Amid Power Grid Outages

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***Abstract*—Prolonged power outages pose a significant threat to the continuity of critical healthcare services. Enhancing resilience requires integrating hazard exposures, infrastructure dependencies, and operational constraints into actionable preparedness plans. We present *BlackOps*, a decision-support framework developed with input from healthcare and hazard analytics experts. The system integrates GIS data on facility locations, hazard footprints, power grids, and road networks to assess vulnerability. To bridge the gap between policy and practice, we leverage context-aware Large Language Models (LLMs) to extract facility-specific required autonomy (operational self-sufficiency under grid failure) and metadata from unstructured regulatory documents. Beyond regulatory compliance, BlackOps quantifies preparedness through indices that measure a facility's ability to replenish resources from regional partners, while accounting for stochastic grid failures and structural damage. These metrics are aggregated into a unified risk score and translated into facility scorecards. These outputs provide decision makers with priority tiers and actionable insights, such as supplemental energy requirements and optimal resource pre-staging locations. Experiments across thousands of scenarios demonstrate the framework's ability to enhance healthcare resilience during multi-day outages.**

***Index Terms*—Power grid outages, GIS, Disaster preparedness, Healthcare resilience, Risk analysis**

I. Introduction

Power grid outages are becoming more frequent, longer, and broader in geographic scope. They are driven by extreme weather and aging infrastructure, and are often amplified by cascading interdependencies across critical services [1]. According to the US Department of Energy (DOE), the leading cause of power outages is weather-related [2], including wildfires, hurricanes, flooding, and more; such outages have increased by nearly 80% in the last decade [3]. Healthcare is among the most power-dependent sectors. Thus, disruption in supply creates an immediate threat to critical life-safety systems. Facilities such as hospitals, skilled nursing, assisted living communities, and dialysis centers rely on continuous power for life-sustaining medical devices, maintenance of blood and pharmacy supplies, and the digital infrastructure required for real-time patient monitoring.

Empirical studies of major disasters have established a direct correlation between power loss and systemic healthcare failures, including backup generator malfunctions, forced facility evacuations, and significant spikes in excess mortality and morbidity among vulnerable hospital and dialysis populations [4], [5]. Events such as the evacuation of hundreds of patients from a Los Angeles hospital after generator failures during Tropical Storm Hilary [6] show how local equipment failures can rapidly escalate into a regional public health crisis.

Prior disaster resilience research (CareDEX) [7], [8] involved wildfire, earthquake, and water-contamination drills and tabletop exercises with partner senior care facilities and first responders. These exercises consistently identified a critical regional planning deficit. When outages persist for multiple days, the primary limiting factor is not the mere existence of backup hardware. Instead, resilience depends on a facility's functional autonomy, which in this work denotes operational self-sufficiency, i.e., the ability to sustain essential services using on-site resources and coordinate resupply amidst degraded transportation networks and uncertain outage durations. Although Hazard Vulnerability Analyses (HVA) and Geographic Information Systems (GIS) provide baseline exposure indicators [9], [10], preparedness data remains fragmented across multiple organizations. Critical but disparate information is distributed through various standards, including those from the National Fire Protection Association (NFPA), the Federal Emergency Management Agency (FEMA), and the Administration for Strategic Preparedness and Response (ASPR TRACIE).

Regional planning is also hindered by the complexity of reconciling fine-grained facility data with site-specific hazards, infrastructure networks, and fragmented regulatory guidance. Thus, current practices struggle to provide actionable insights required by emergency operations centers (EOCs) to identify which facilities will exhaust their autonomy or where resupply efforts will be most constrained. In most jurisdictions, these resources remain ambiguous: GIS dashboards provide spatial context, while facility checklists and guidance remain document-based, often via spreadsheets. This makes regional planning difficult [11], [12]. This paper presents *BlackOps*, a regional preparedness framework that uses spatial analytics to assess healthcare risk during prolonged power outages. Developed with input from local government officials, including Orange County Health Care Agency (OCHCA), healthcare practitioners, and disaster resilience experts (ImageCat, FPRF/NFPA), the framework aligns with real-world operational workflows and governance concerns. Our key contributions include:

- A novel GIS-based risk formulation specifically designed to quantify healthcare preparedness.

- Comprehensive vulnerability indices that utilize GIS data with physics-based infrastructure models to capture the multi-dimensional complexities of real-world outages.
- A workflow leveraging Large Language Models (LLMs) to transform unstructured healthcare power-related guidelines into queryable JSON schemas.
- Emulation-based scenarios to identify vulnerable facility clusters and provide actionable situational awareness.

This paper is organized as follows. §II provides background on regional preparedness and an overview of the BlackOps framework. We model the problem in §III, compute risk indices in §IV, and provide the overall system in §V. Then, experiments and future directions are discussed in §VI and §VII.

II. Background and Overview

Background and Challenges: Regional-scale preparedness typically combines spatial data with facility-level assessments using frameworks such as FEMA's Threat and Hazard Identification and Risk Assessment (THIRA) [11]. Tools such as Hazus model hazard impacts [9], while indices like the CDC SVI [10] and other resilience frameworks [8], [13], [14] incorporate social context. From an infrastructure perspective, grid resilience studies use topology-aware metrics to identify vulnerabilities and estimate disruption [15], [16], remote sensing to quantify exposure [17], [18], and network modeling to characterize hospital access during disruptions [19]. However, during prolonged failures, it is difficult to synthesize these methods into the site-specific indicators needed to model backup autonomy. Along with GIS analytics, healthcare facilities use readiness instruments to structure emergency planning. While HVA tools assign ordinal scores to the probability and impact for a range of hazard events [20], the ASPR Risk Identification and Site Criticality (RISC) Toolkit uses customizable scoring and guided assessments to evaluate facility reliance on external utilities [12]. This typically produces spreadsheet-oriented outputs that lack the spatial context needed during disasters. Transforming regulatory text into computable representations has also evolved from curated logic engines [21] to semantic parsing and relation extraction [22]. However, these techniques fail to connect linguistic extraction with geospatial modeling: regulatory requirements are not mapped to the spatial attributes and operational timelines needed for evaluating resilience. Automated extraction requires rigorous validation before integration into high-stakes emergency management.

Key limitations: Current practices yield fragmented views of outage preparedness. Qualitative instruments create an objective gap where ordinal scores are not easily comparable across facilities, and disconnects between regulatory text extraction and geospatial layers prevent the integration of policy requirements with physical infrastructure models. These shortcomings motivate a GIS-informed risk framework that synthesizes disparate data into actionable insights.

To address these limitations, BlackOps integrates three primary data sources: (i) physical infrastructure data describing healthcare facilities, electric-grid components, and hazards; (ii) emergency-power regulatory documents processed

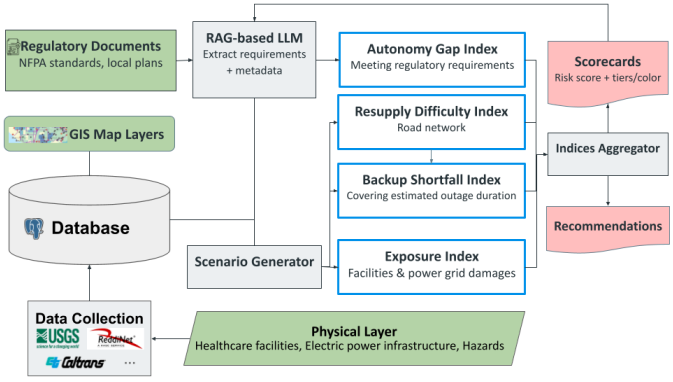


Figure 1. *The BlackOps Architecture*

using retrieval-augmented generation (RAG) to extract facility-relevant autonomy requirements and context-aware metadata; and (iii) GIS layers that provide published spatial services such as historical traffic patterns, typical peak-hour congestion data, and the Social Vulnerability Index (SVI) to model the socio-economic context. Scenarios are instantiated from the USGS National Seismic Hazard Model Project [23] and simulated through the NGA-West2 ground motion simulator [24]. BlackOps computes a set of interpretable indices including static measures (social vulnerability and autonomy gaps) and scenario-dependent measures (routing-based resupply difficulty, backup shortfall, and exposure) as defined in §IV. These indices are then combined by an aggregator into a unified risk score and translated into per-facility scorecards and recommendations. Scorecards summarize a facility's overall vulnerability, while recommendations provide derived, actionable planning guidelines for decision-makers and stakeholders. The overview of the BlackOps approach is shown in Fig. 1.

III. Problem Formulation

This paper aims to answer the question: ***Given a prolonged power outage scenario in a region, which health care facilities are most at risk of losing critical services, and how should planners reduce that risk?*** Inputs include hazard and grid outage scenarios, a catalog of health care facilities with backup-power characteristics, transportation-network information for resupply, and requirements from emergency power regulatory standards. For each facility and scenario, we compute a set of interpretable indices that quantify exposure, backup shortfall, resupply difficulty, autonomy gaps, and social vulnerability, which enables regional preparedness assessment. While social vulnerability v_f and the autonomy gap g_f indices are static components, the remaining indices are scenario-dependent (described later in §IV-B).

Let $\mathcal{F}$ be the set of health care facilities, indexed by f with attribute vector a_f, capturing its type, location, estimated critical electrical demand, generator capacity, usable fuel or battery storage, operational status, etc. $\mathcal{S}$ is the set of outage scenarios where each $s \in S$ represents a stochastic earthquake event and its induced damage state including expected grid-outage duration $T^{\text{out}}_{f,s} \geq 0$ (hours), with $T^{\text{out}}_{f,s}=0$ when grid power remains available at f. Let B denote the set of electric

substations. We use $z_{b,s} \in \{0,1\}$ to indicate whether substation b remains operational in scenario s. In addition, $\pi(f) \in B$ maps facility f to its serving substation. We define the road network as a graph $G = (V, E)$. For each road segment $e \in E$, $\kappa_{e,s} \in \{0,1\}$ indicates whether e is traversable in scenario s.

During multi-day outages, autonomy can be extended if resupply begins early enough and routes remain usable. Therefore, for each scenario s, we estimate the earliest feasible resupply start time $T^{\text{start}}_{f,s} \geq 0$ (hours) by combining shortest-path travel times on the road network with scenario-specific closures (blocked roads) and historical traffic information [25]. We compute route redundancy $\rho_{f,s} \in [0,1]$ from the road network graph as a normalized count of K-shortest paths from supply depots to f under scenario s, such that $\rho_{f,s} \approx 0$ indicates a bottleneck route, while larger values indicate multiple alternatives. To quantify the operational challenge of restoring resources to a facility, we introduce a ***resupply difficulty index***, $d_{f,s} \in [0,1]$. This captures the logistical strain caused by both the timing of the response and the vulnerability of the local transport network. Eq. (1) defines this metric below.

$$d_{f,s} = \begin{cases} 0 & T^{\text{out}}_{f,s}{=}0, \\ \text{clip}_{[0,1]}\Big(\alpha_d \frac{T^{\text{start}}_{f,s}}{T^{\text{out}}_{f,s}} + (1{-}\alpha_d)(1{-}\frac{\rho_{f,s}}{K})\Big) & T^{\text{out}}_{f,s}{>}0 \end{cases} \quad (1)$$

Here, $\alpha_d \in [0,1]$ weights the relative importance of operational bottlenecks: $\alpha_d \approx 1$ prioritizes timeliness (i.e., how late resupply begins relative to the outage), while $\alpha_d \approx 0$ prioritizes the availability of unblocked K-shortest paths from supply origins to facility f in scenario s. This formulation makes $d_{f,s}$ interpretable and tunable for jurisdictions that view response speed versus network robustness as a key constraint.

In addition, we express how much of the outage horizon can be covered by on-site energy plus feasible deliveries as the effective autonomy. We define this in Eq. (2) as follows:

$$T^{\text{eff}}_{f,s} = T^{\text{crit}}_{f} + E^{\text{del}}_{f,s}/P^{\text{crit}}_{f} \quad (2)$$

where $E^{\text{del}}_{f,s} \geq 0$ is the delivered usable energy (kWh) that can reach f under scenario s within the outage window, given access and resource availability assumptions. Therefore, the ***shortfall index***, as depicted in Eq. (3) is the fraction of outage duration *not covered* by effective autonomy:

$$s_{f,s} = \begin{cases} 0, & T^{\text{out}}_{f,s} = 0, \\ \text{clip}_{[0,1]}\left(\frac{T^{\text{out}}_{f,s} - T^{\text{eff}}_{f,s}}{T^{\text{out}}_{f,s}}\right), & T^{\text{out}}_{f,s} > 0, \end{cases} \quad (3)$$

with $s_{f,s} = 0$ indicating full coverage of the outage horizon, and larger values indicate expected loss of critical services. Furthermore, ***exposure index*** $e_{f,s} \in [0,1]$ captures how strongly scenario s is expected to impact facility f (see §IV-B3). This is derived from established GIS-based hazard models, including shaking intensity, fragility curves for earthquakes, and power grid feeders vulnerability [26]. Hence, for each facility-scenario pair (f, s), we aggregate indices into an overall preparedness risk score as follows:

$$r_{f,s} = \sum_{i=1}^{n} w_i\, \phi_i(f,s), \quad s.t. \sum_{i=1}^{n} w_i = 1, \quad w_i \geq 0, \quad (4)$$

where components are $\phi_i = \{e_{f,s}, s_{f,s}, d_{f,s}, g_f, v_f\}$ and w_i are policy weights set by decision makers. The weighted sum is used as an interpretable planning score rather than a fitted predictor. Hence, all components are normalized to $[0,1]$, so each term contributes a comparable marginal risk. Equal weights provide a neutral default when local priorities are unavailable. During deployment, w_i can be tuned through expert elicitation or sensitivity analysis. For example, regions with severe transit constraints may emphasize $d_{f,s}$, while local SVI availability may motivate reweighting. Under this framework, the most vulnerable facility is identified as:

$$R^{\max}_{s} = \max_{f \in \mathcal{F}} r_{f,s} \quad (5)$$

Finally, to measure regional operational efficiency, we cluster facilities into C_s risk tiers as shown in Eq. (6). Here, $\mu_{c,s}$ is the centroid risk score for tier c in scenario s and $c_{f,s}$ is the tier assignment.

$$\min_{\{c_{f,s}\},\{\mu_{c,s}\}} \sum_{f \in \mathcal{F}} \left(r_{f,s} - \mu_{c_{f,s}}\right)^2, \quad c_{f,s} \in \{1, \ldots, C_s\}. \quad (6)$$

IV. Modeling Preparedness Indices

In this section, we illustrate how we model the indices used for assessing the vulnerability of healthcare facilities.

A. Modeling Static Indices for Communities

1) Social Vulnerability Index (SVI): Risk is typically modeled as a function of hazard, vulnerability, and available resources [27]. While hazard is an external factor, vulnerability can be modeled as a pre-existing community condition that can amplify disaster impacts. The CDC uses a percentile ranking $[0,1]$ derived from 16 U.S. Census indicators aggregated into four themes: socioeconomic status, household composition/disability, minority status/language, and housing/transportation [10]. Its computation follows a hierarchical ranking pipeline where raw indicators are converted to percentiles, aggregated, and re-ranked into themes. This is then combined into a composite regional score [27]. We map healthcare facilities to these scores by overlaying their coordinates onto the SVI layer; this mapping is presented later in our experiments in Fig. 4. Hence, smaller facilities can exhibit higher vulnerability than major centers. However, SVI alone does not capture a facility's ability to maintain critical services during grid failure. To characterize vulnerability to prolonged power interruptions, we combine this with operational metrics.

2) Autonomy Gap Index: Prolonged outages threaten continuity of care when on-site energy reserves are exhausted before utility restoration. We quantify this as a temporal deficit when a facility's estimated **critical autonomy** T^{crit}_{f} is less than the **required autonomy** T^{req}_{f} prescribed by regulatory and accrediting guidance (e.g., NFPA, FEMA, or CMS).

Characterizing critical autonomy (T^{crit}_{f}): This is derived from facility maintenance records, generator logs, and Emergency Operations Plans (EOPs) (reviewed and validated through drills and tabletop exercises). If $P^{\text{crit}}_{f}{>}0$ represents the critical electrical power demand (kW) required to sustain life-safety and essential branches during an outage, and $E^{\text{store}}_{f} \geq 0$

Algorithm 1: RAG-based Extraction of Requirements

Input: Vector database $\mathcal{V}$, Facility attributes a_f, Parameter k
Output: Required autonomy T_f^{req}, Metadata M_f

// STEP 1: Convert attributes into strings and embed
$s_q, v_q \leftarrow \text{StringifyAndEmbed}(a_f)$
// STEP 2: Search database for the best matches
$Scores \leftarrow [\,]$
for $c \in \mathcal{V}$ **do**
  $sim \leftarrow \text{CosineSimilarity}(v_q, c.\text{emb})$
  $Scores.\text{append}((sim, c))$
// STEP 3: Retrieve top-k (most relevant) chunks
Sort $Scores$ descending by sim
$Context \leftarrow \emptyset$
for $i \leftarrow 1$ **to** k **do**
  $(sim_i, c_{top}) \leftarrow Scores[i]$
  $Context \leftarrow Context \oplus c_{top}.\text{text}$
// STEP 4: Construct prompt and run inference
$\rho \leftarrow \text{Instruction}() \oplus Context \oplus s_q$
$J \leftarrow \text{LLM}(\rho, \text{mode} = \text{JSON})$
// STEP 5: Parse and validate output
if *isValidNumber(J.hours)* **then**
  $T_f^{\text{req}}, M_f \leftarrow J.\text{hours}, \{J.\text{condition}, J.\text{doc_id}\}$
else $T_f^{\text{req}}, M_f \leftarrow \text{FlagManualReview}()$
return T_f^{req}, M_f

denotes usable on-site energy (kWh) available from backup resources, then the critical autonomy is obtained as follows:

$$T_f^{\text{crit}} = E_f^{\text{store}} / P_f^{\text{crit}} \tag{7}$$

Hence a facility with 100,800 kWh of fuel energy (E_f^{store}) and a critical demand load (P_f^{crit}) of 1,400 kW has a critical autonomy of T_f^{crit}=72 hours. Per NFPA safety standards [28], demand estimates incorporate a 10% to 25% safety margin to account for unexpected surges that occur during disasters.

Extracting required autonomy (T_f^{req}): This is challenging because requirements are scattered across heterogeneous federal, state, and local documents. We address this using Large Language Models (LLMs) integrated with Retrieval-Augmented Generation (RAG) to mitigate hallucinations and ensure regulatory grounding (Alg. 1). The extraction follows a five-step pipeline. First, facility attributes a_f are stringified and embedded into a high-dimensional vector space (Step 1). We perform a similarity search against a vector database $\mathcal{V}$ of regulatory codes, using cosine similarity to identify the k most relevant chunks (Steps 2–3). These chunks are appended to a structured instruction prompt ρ, which directs the LLM to perform inference and return results in a JSON format (Step 4). Finally, the system validates the output; if a valid numeric value is found, it is returned as T_f^{req} along with its citation metadata M_f. Otherwise, it is flagged for manual review (Step 5). This yields the autonomy gap index g_f:

$$g_f = \text{clip}_{[0,1]}\left(\frac{T_f^{\text{req}} - T_f^{\text{crit}}}{T_f^{\text{req}}}\right) \tag{8}$$

Hence, a facility with T_f^{req}=96 hours but maintaining only T_f^{crit}=72 hours, has g_f=0.25, which indicates a 25% deficit.

B. Modeling Scenario-dependent Indices

1) Assessing the Resupply Difficulty Index: A facility's preparedness also depends on its ability to replenish exhausted resources from external partners. While on-site autonomy is finite, it can be extended if external resupply begins early and transport routes remain usable. We quantify this logistical challenge through the resupply difficulty index $d_{f,s}$ (Eq. (1)), which balances temporal delays against structural network fragility. This metric shifts the focus from static building attributes to the dynamic strain caused by response timing and transport network vulnerability. To model the infrastructure, we use the ArcGIS network analyst extension [25] to represent the road system as a weighted graph where edges incorporate historical traffic data to account for typical congestion. Health care facilities typically maintain partnerships with other sister facilities (within the same network) or contracted vendors to secure resources during emergencies [8]. Consequently, we evaluate the earliest feasible resupply start time $T_{f,s}^{\text{start}}$ using Dijkstra's algorithm to identify a set of K potential shortest paths between a facility f and its suppliers.

We define the route redundancy $\rho_{f,s}$ as the specific count of these K paths that remain accessible after removing edges to reflect scenario-specific road closures (detailed in §IV-B3). As defined in Eq. (1), the difficulty index captures two primary operational bottlenecks. *Temporal constraints*: higher resupply start time $T_{f,s}^{\text{start}}$ relative to the outage duration increases $d_{f,s}$; *Structural constraints*: difficulty is inversely related to route availability, i.e., as the number of accessible partners drops toward zero, the term $(1 - \rho_{f,s}/K)$ approaches one, signaling a near single point of failure. We use α_d as a tunable knob to prioritize the timeliness of delivery and the availability of access routes. Ultimately, this index provides a comprehensive measure of the logistical constraints faced by facilities during a disaster. This index is critical because it dictates the feasibility and timing of resource arrivals, which in turn determines the total energy supply available to mitigate the electric power demand to cover the estimated duration of the outage.

2) Quantifying the Backup Shortfall Index: Defined as $s_{f,s} \in [0, 1]$, this index measures the fraction of an outage duration not covered by a facility's total accessible power. A value of $s_{f,s} = 0$ indicates full coverage, while higher values represent periods where critical services are projected to fail due to energy exhaustion. A facility's effective autonomy is thus the duration it can sustain operations by combining its static on-site capacity with dynamic replenishment resources. Per Eq. (2), it is computed as $T_{f,s}^{\text{eff}} = T_f^{\text{crit}} + (E_{f,s}^{\text{del}} / P_f^{\text{crit}})$, indicating the combined critical autonomy (minimum to sustain critical equipment and services) and the autonomy produced by the electrical backup equipment delivered by the partnered facilities. The expected outage duration $T_{f,s}^{\text{out}}$ is derived using FEMA Hazus-MH's methodology, which estimates restoration timelines by applying fragility curves to utility infrastructure (substations and distribution nodes) under scenario-specific hazard intensities (see §IV-B3). Consider a facility with P_f^{crit}=90 kW during a 72-hour outage ($T_{f,s}^{\text{out}}$=72). With

RAG Prompt

```
Task:Extract mandated backup hours using ONLY the excerpts
Excerpts:<relevant vector DB passages>
Return:{"hours":<num>,"condition":<text>,"doc_id":<id>}|"unknown"
Rule:If hours are not found in the excerpts,return "unknown"(no guessing)
```

only 24 hours of critical autonomy (T_f^{crit}=24), it requires 48 hours of supplemental power, necessitating a delivery of $E_{f,s}^{\text{del}} = 4,320$ kWh. If logistical difficulty ($d_{f,s}$) limits the feasible delivery to only 2,160 kWh (providing 24 hours of power), the effective autonomy reaches only 48 hours (24 + 24). Therefore, we quantify this facility's shortfall as $s_{f,s}$=(72 − 48)/72 = 0.33.

3) Physics-Based Hazard Exposure and Damage Estimation: In Alg. 2, we detail the process of capturing the total damage and loss of function of facilities caused by the hazard. This requires modeling the initial damage (lines 2-7) and its resulting downstream impacts (lines 8-12). To keep the paper focused, we consider earthquake hazard events.

We start by leveraging historical catalogs of hazard events to instantiate and run N stochastic simulations. The likelihood of a hazard scenario depends on probability distributions that describe where such events are likely to occur, e.g., proximity to fault lines. For earthquake events, this data can be obtained from USGS [23], and simulated using NGA-West2 [29]; this estimates aspects such as the scale and intensity of ground motion, site effects, and epistemic uncertainty. We then analyze each hazard event for its potential to cause physical damage and loss of function to both facilities and electric grid components using FEMA Hazus [30]. This tool uses engineering-based fragility and functionality models to estimate structural damage, service disruption, and downtime. This yields a *damage ratio* and *loss of function* value for each facility f and grid substation x, as the extent of damage and subsequent reduction in operational capacity, respectively.

Following substation failures, the grid reroutes power subject to power flow balance and capacity constraints, but may need to shed load to maintain safe operation. This leads to an increased loss of function in nearby facilities. We extend our prior physics-based grid model [26] to compute post-hazard flows in Eq. (9). When a substation's damage ratio exceeds a threshold τ, we assume that it is no longer safe to operate. Such failures propagate through the network, i.e., starting with the power lines connected to the substation. To capture the cascading failure, we relate several power grid concepts[1]: generation p^g, demand p^d, load shed p^s, flows p, maximum generation $\bar{p}^g$, line capacity t, reactance X, and phase angles θ to the failures x.

$$\max \quad \sum_{i\in\mathcal{N}}(p_i^d - p_i^s) \tag{9a}$$

$$\text{s.t.} \quad p_i^g - p_i^d + p_i^s = \sum_{(i,j)\in\mathcal{E}^f(i)} p_{i,j} - \sum_{(j,i)\in\mathcal{E}^t(i)} p_{j,i} \quad \forall i \in \mathcal{N} \tag{9b}$$

$$p_{i,j} = X_{ij}^{-1}(\theta_i - \theta_j)(1 - x_{i,j}) \quad \forall (i,j) \in \mathcal{E} \tag{9c}$$

$$0 \le p_i^g \le (1 - x_i)\bar{p}_i^g \quad \forall i \in \mathcal{N} \tag{9d}$$

$$-t_{i,j}(1-x_{i,j}) \le p_{i,j} \le t_{i,j}(1-x_{i,j}) \forall (i,j) \in \mathcal{E} \tag{9e}$$

The objective in (9a) maximizes served power in the network, subject to power flow physics (9b, 9c) and grid capacity limits (9d, 9e). Solving this optimization problem can then be accomplished by using linear program solvers [31] offline.

[1]Subscripts are omitted for ease of presentation

Algorithm 2: Computing Physics-based Damage and Loss of Function

Input: Hazard Event Catalog $catalog$, Community cm, Electric Grid $\mathcal{G}$, Facilities $\mathcal{F}$, int N
Output: Damage States $\mathcal{D}$, Estimated Loss of Function $\mathcal{L}$

1 $\mathcal{D} \leftarrow \{\}$ $\mathcal{L} \leftarrow \{\}$ // Damage, Loss of function repository
2 **for** $i \leftarrow 1$ **to** N **do** // Phase 1: Model Initial Damage
3 $\quad s \leftarrow InitializeHazard(catalog, cm)$
4 $\quad res \leftarrow SimulateHazard(s)$
5 $\quad damage, loss \leftarrow EstimateDamageAndLoF(res, \mathcal{F})$
6 $\quad \mathcal{D}.addDamageRatioScenario(s, damage)$
7 $\quad \mathcal{L}.addLossOfFunctionScenario(s, loss)$
8 **for** $s \leftarrow \mathcal{S}$ **do** // Phase 2: Model Cascading Impact
9 $\quad X \leftarrow GetHighlyDamagedComponents(s)$
10 $\quad \mathcal{G}^* \leftarrow ExtractReducedGrid(\mathcal{G}, X)$
11 $\quad \{p_i^s\} \leftarrow$ Solve for power flow in (9)
12 $\quad \mathcal{L}.updateLossOfFunction(s, \{p_i^s\})$
13 **return** $\mathcal{D}, \mathcal{L}$

V. Towards Facility Preparedness Scorecards and Recommendations

Overall, we synthesize the individual indices into a unified preparedness risk score, denoted as $r_{f,s} \in [0, 1]$. This composite score is computed by aggregating all the indices via a weighted sum as defined in Eq. 4 (lines 2–12 in Alg. 3). Based on Eq. 5, $R_s^{\max}$ represents the most vulnerable facility in each scenario s. However, for operational efficiency, we partition facilities into C_s discrete risk tiers. We implement this using k-means clustering by grouping facilities into risk tiers based on the proximity of their risk scores to computed tier centroids (line 13). We record the average risk score $\frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} r_{f,s}$ (line 15) and the worst-case risk $\max_{s\in\mathcal{S}} r_{f,s}$ and map the average score to a tier/color as shown in Table I. Tier labeling and interpretations are inspired by the Simple Triage And Rapid Treatment (START) convention [32], where tiers use fixed risk-score ranges for consistent interpretation across scenarios (line 16). Because we evaluate the risk score through many outage scenarios (∼1000), the *scorecard* is reported per facility by summarizing risk over the scenario ensemble.

BlackOps also supports the following recommendations: #1) The optimal supplier(s) selected by minimizing the resupply difficulty ($d_{f,s}$) (based on top K suppliers). #2) The amount of power needed to cover the estimated deficit. Here, the deficit-hours x correspond to the unmet autonomy duration $max(0, T_f^{\text{req}} - T_f^{\text{crit}})$ and P_f^{crit} indicates the facility's critical power demand. #3) The LLM snapshot summarizes the relevant excerpts discussed in §IV-A2. Note that the snapshot does not affect the risk score, but rather serves as a context-aware summary of the facility.

Scorecard – **Facility:** *[name, type, location]*
Risk score (avg): $\frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} r_{f,s}$
Risk score (max): $\max_{s\in\mathcal{S}} r_{f,s}$
Tier/Color: *[Tier, Color from Table I]*
Interpretation: *[row meaning from Table I]*
Recommendations: #1) Optimal supplier(s): *[facility/vendor]* #2) Based on x hrs autonomy deficit ⇒ provision ≈ $x \times P_f^{crit}$ kWh usable backup energy. #3) LLM snapshot: *["context-aware summary of facility f"]*

Algorithm 3: Scorecards & Recommendations

Input: Facilities $\mathcal{F}$, Scenarios $\mathcal{S}$, GIS layers $\mathcal{L}$, Requirements $\{T_f^{\text{req}}, M_f\}$, Number of tiers C_s
Output: Risk scores $\{r_{f,s}\}$, Tiers $\{c_{f,s}\}$, Scorecards SC, Recommendations REC

$\mathsf{SC} \leftarrow \{\}$; $\mathsf{REC} \leftarrow \{\}$ // scorecard, recommendation
for $f \in \mathcal{F}$ **do**
  $T_f^{\text{crit}} \leftarrow \textsc{CriticalAutonomy}(P_f^{\text{crit}}, E_f^{\text{store}})$
  $g_f \leftarrow \textsc{AutonomyGap}(T_f^{\text{req}}, T_f^{\text{crit}})$
  // $\mathcal{L}(\cdot)$ are queryable GIS layers (feature class (point, line, polygon)/raster/network dataset)
  $v_f \leftarrow \textsc{SVIJoin}((\text{lat}_f, \text{lon}_f), \mathcal{L}.\text{SVI})$
for $s \in \mathcal{S}$ **do**
  $(H_s, \{T_{f,s}^{\text{out}}\}) \leftarrow \textsc{ScenarioGen}(\mathcal{L}.\text{Hazards}, s)$
  for $f \in \mathcal{F}$ **do**
    $e_{f,s} \leftarrow \textsc{Exposure}((\text{lat}_f, \text{lon}_f), H_s, \mathcal{L}.\text{Grid})$
    $d_{f,s} \leftarrow \textsc{ResupplyDiff}((\text{lat}_f, \text{lon}_f), H_s, \mathcal{L}.\text{Roads})$
    $s_{f,s} \leftarrow \textsc{Shortfall}(T_f^{\text{crit}}, P_f^{\text{crit}}, T_{f,s}^{\text{out}}, d_{f,s})$
    $r_{f,s} \leftarrow \textsc{RiskScore}(e_{f,s}, s_{f,s}, d_{f,s}, g_f, v_f, \{w_i\})$
  $\{c_{f,s}\}_{f \in \mathcal{F}} \leftarrow \textsc{KMeansCluster}(\{r_{f,s}\}_{f \in \mathcal{F}}, C_s)$
for $f \in \mathcal{F}$ **do**
  $\bar{r}_f \leftarrow \textsc{Mean}(\{r_{f,s}\}_{s \in \mathcal{S}})$
  $(\text{Tier}_f, \text{Color}_f, \text{Interp}_f) \leftarrow \textsc{TierMap}(\bar{r}_f)$
  $\Delta E_f \leftarrow \textsc{EnergyDeficit}(T_f^{\text{req}}, T_f^{\text{crit}}, P_f^{\text{crit}})$
  // Closest accessible supplier(s)
  $\text{Sup}_f \leftarrow \textsc{OptSupplier}(f, \{d_{f,s}\}_{s \in \mathcal{S}}, \mathcal{L}.\text{Suppliers})$
  $\text{Snap}_f \leftarrow \text{LLM}(M_f)$ // LLM snapshot from metadata
  $\mathsf{SC}[f] \leftarrow (\bar{r}_f,\ \text{Tier}_f,\ \text{Color}_f,\ \text{Interp}_f)$
  $\mathsf{REC}[f] \leftarrow (\text{Sup}_f,\ \Delta E_f,\ \text{Snap}_f)$
return $\{r_{f,s}\}, \{c_{f,s}\}, \mathsf{SC}, \mathsf{REC}$

Table I
FACILITY RISK TIERS

$r_{f,s}$	Tier	Description
$[0, 0.25)$	Low (Green)	Stable: minimal external action expected.
$[0.25, 0.50)$	Moderate (Yellow)	Monitor: secondary actions as needed.
$[0.50, 0.75)$	High (Orange)	Urgent: resources likely needed mid-outage.
$[0.75, 1]$	Extreme (Red)	Critical: immediate resource/pre-staging needed.

VI. EXPERIMENTS

Setup: We evaluated BlackOps using stochastic earthquake events from the USGS 2018 NSHMP catalog [23], and estimated ground motion using NGA-West2 [24]. Damage analysis then applied the FEMA Hazus Earthquake Model [30]. Facility data (974 sites) were obtained from the California Department of Public Health (CDPH) [33], and power grid infrastructure data was sourced from the California Test System (CATS) [34]. We stored these in a PostgreSQL/PostGIS database to enable efficient geospatial queries. Road network data and historical traffic information were integrated via ArcGIS Pro to model distances and travel times between facilities. Requirements and metadata were extracted via a distributed client-server architecture, offloading inference to a workstation (AMD Ryzen7 5800X, NVIDIA RTX3080 GPU). Using the Ollama API [35], we deployed GPT-OSS-20b and Gemma 3 12B models with 32k context windows.

To validate the LLM-based extraction, we ran a small sanity check with curated excerpts from NFPA 99 and 110 standards. We evaluated (i) extraction accuracy by querying for requirements with known T_f^{req} values in the excerpts (e.g., 96 hours) and (ii) hallucination resistance using negative-control queries whose requirements were absent from the provided text. The model was required to return "unknown" and trigger manual review rather than invent values. This provides a bounded grounding check; broader LLM validation remains future work. The next set of experiments consists of long-term seismic risk quantification and spatial damage characterization, resupply routing optimization, end-to-end risk integration, and comparative benchmarking against an existing federal assessment tool (ASPR RISC 2.0).

We quantified long-horizon seismic risk using the "10% in 50 years" exceedance baseline. Fig. 2(a) shows the structural damage probability $P(D{>}d)$, where the curve identifies the fraction of facilities exceeding critical thresholds. We then estimate the probability $P(T_{\text{out}}{>}t)$ that facilities will exceed specific outage durations (Fig. 2(b)). This provides a predictive stress-test for response timelines by highlighting a "prolonged failure tail" where estimated recovery durations (t) significantly exceed the standard 96-hour backup autonomy.

In Fig. 3, we illustrate spatial damage results for Orange County facilities. Damage severity is classified based on the damage ratio d: None ($<1\%$), Slight (1-6%), Moderate (6-30%), Extensive (30-75%), and Complete ($>75\%$). The high-quantile damage (Q90) captures high-impact outcomes, indicating the level of exposure a facility experiences in approximately 1 out of 10 scenarios. In contrast, the mean damage summarizes the typical (central-tendency) impact across the scenario ensemble. With these marginal metrics, we compute the tail–mean damage differential (Δ), which quantifies cross-scenario dispersion and highlights facilities whose tail-risk is disproportionately larger than their typical outcome. Finally, the Getis–Ord G_i^* identifies statistically significant spatial correlation where high (or low) damage forms non-random hot/cold clusters (computed via z-scores and confidence levels). The G_i^* statistic distinguishes isolated outliers from statistically significant clusters by evaluating each facility in the context of its neighboring values.

We also use SVI to prioritize resource acquisition from low-vulnerability areas, thereby minimizing the burden on highly vulnerable communities. Travel times between origins and top-ranked partners (Fig. 4) are computed using Dijkstra's shortest-path algorithm, which incorporates historical traffic data to account for congestion (at different times of the day). To simulate infrastructure degradation, road segments within a defined buffer radius of facilities and substations reaching 'Extensive' or 'Complete' damage states are assumed to be impassable. Furthermore, to optimize the routing search space, we prioritize higher-order road networks. Hence, the resupply

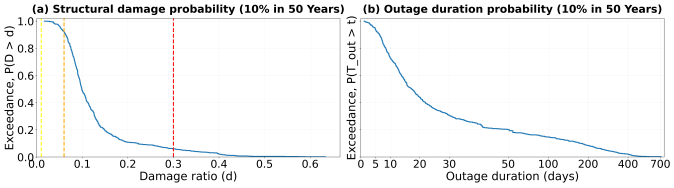


Figure 2. *Exceedance Curves (Long-term Earthquake Risk)*

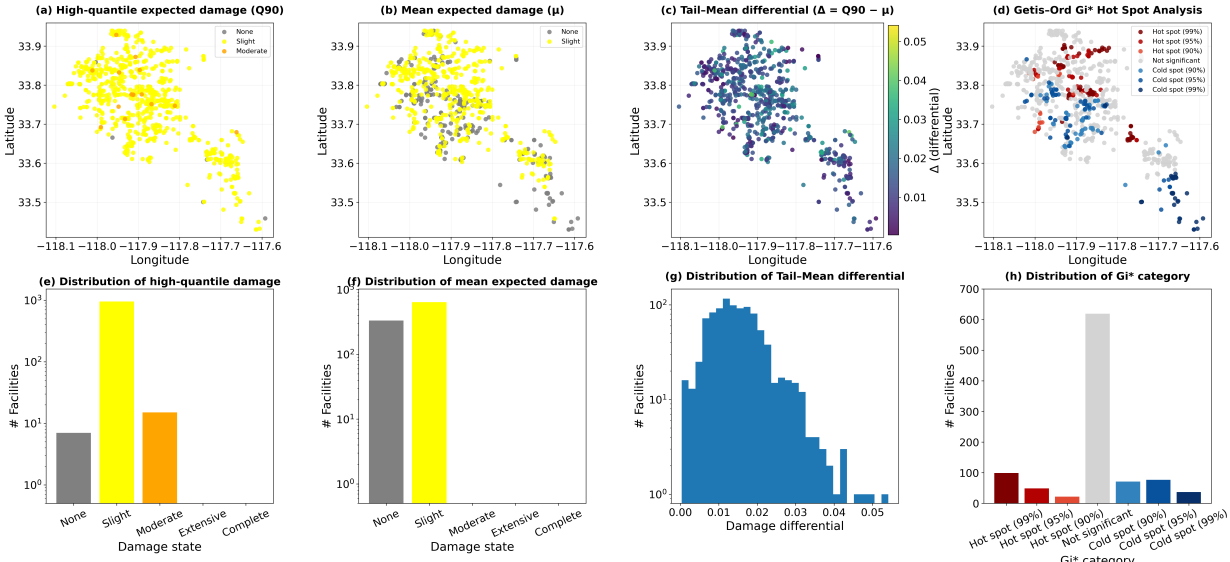


Figure 3. *High-quantile Damage (Q90), Mean Damage (μ), Tail–mean Damage Differential (Δ), Getis–Ord G_i^* Hot/cold Spots with Distributions*

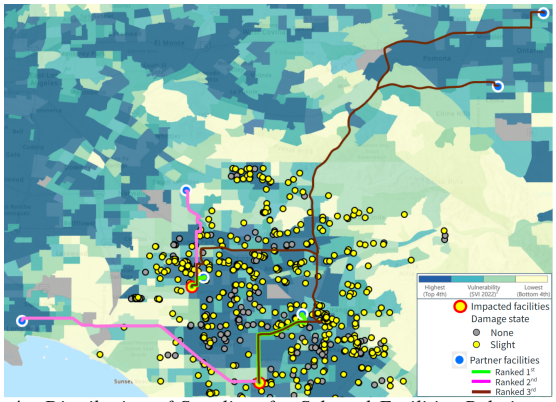


Figure 4. *Distribution of Suppliers for Selected Facilities Relative to Social Vulnerability Index (SVI)*

difficulty index d_f is obtained by mapping each facility's top-K partner set to the earliest feasible start time T_f^{start} and the availability count ρ_f capturing how many partners are reachable. For one of the illustrative facilities in Fig. 4 with $K{=}5$ candidate suppliers, we obtain $d_f{=}0.240$ with $T_f^{\text{start}}{=}3.42\,\text{h}$, and $\rho_f{=}3$ corresponding to $9.03/45.46/84.00\,\text{km}$.

Fig. 5 provides end-to-end results of the overall risk score after combining all indices. As shown in Fig. 5(a), the overall risk distribution reveals a non-linear gradient with a steep, small subset of high-priority facilities (left), whereas the long, flatter tail indicates that the majority of the network clusters within moderate-to-lower risk levels. This is further quantified by the risk exceedance (Fig. 5(b)) and the resulting vulnerability classification (Fig. 5(c)), which together provide a visual breakdown of how many facilities fall into high-risk versus moderate-risk tiers. However, the most critical insights emerge from the indices comparison (Fig. 5(d)), where it is evident that the top-100 high-risk facilities exhibit significantly higher mean intensities across all dimensions. This difference is most notable in the shortfall, SVI, and exposure indices.

To validate the integration of all indices, we conducted an ablation study as shown in Fig. 6. The sensitivity analysis defines the operational bounds of the model by mapping the minimum and maximum possible risk scores across all indices (Fig. 6(a)). The shaded grey area (the envelope) represents the sensitivity of our model. Hence, wide shading indicates where the facility's rank is highly dependent on a specific index. i.e., a facility could have very high social vulnerability but low structural damage (exposure). Therefore, by dropping the SVI, its risk score will considerably decrease. Similarly, a narrow shading indicates a "unanimous" risk profile where the indices converge (at the very high and very low ends of the rank). Thus, by removing the SVI and exposure indices, we highlight the importance of a multi-dimensional approach, as shown (Fig. 6(b)), which reveals significant miss rates of 20% to 40% when neglecting SVI or exposure. Furthermore, the displacement analysis (Fig. 6(c)) and the high degree of tier disagreement (Fig. 6(d)) confirm that without this integration, nearly 40% of the most critical facilities would be misclassified by decision makers.

In addition, to benchmark against an established federal preparedness framework, we utilize the ASPR RISC Toolkit 2.0 [12] for a representative facility (UCI Health Fountain Valley) and compare its risk ratings to the corresponding BlackOps estimates. RISC, which computes risk as the product of Threat/Hazard, Vulnerability, and Consequence modules (THAM×RIST-V×RIST-C), yielded an earthquake exposure score of 0.96, whereas BlackOps produced an exposure value of 0.67, corresponding to a 30.2% reduction relative to the national-default assessment. In contrast, RISC reported a vulnerability score of 0.02 (national value) for the same facility, while the fine-grained census-tract and facility-specific inputs used in BlackOps resulted in an SVI of 0.76. Because RISC documentation explicitly notes that THAM hazard ratings are derived from national-level data and may be overwritten when local data or subject-matter expertise indicates different likelihoods, we interpret these differences as methodological rather than contradictory, with BlackOps as a systematic local refinement layer that can complement standardized RISC assessments for regional outage planning.

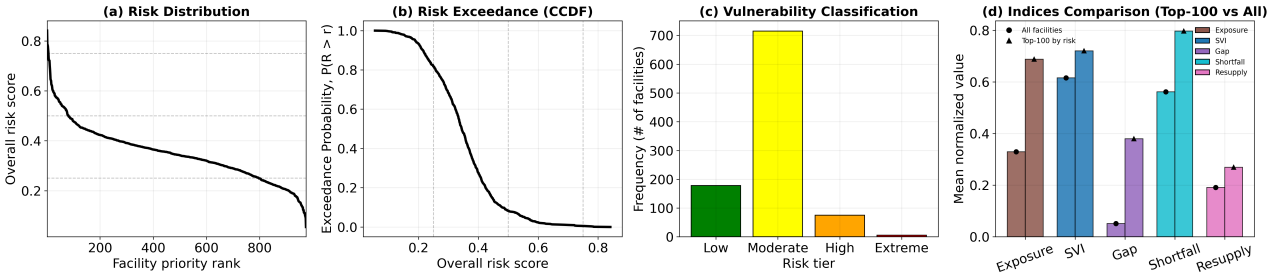


Figure 5. *End-to-End Risk Evaluation*

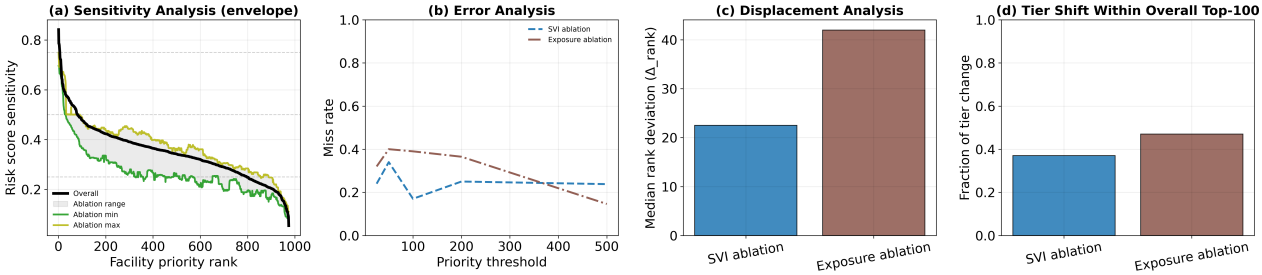


Figure 6. *Risk Model Ablation Study*

VII. Conclusion and Future Work

This paper presents a framework for healthcare resilience amid prolonged power outages. By coalescing hazard footprints, utility dependencies, and operational constraints, we provide a comprehensive assessment of vulnerable facilities. Physics-based simulations conducted with the Orange County healthcare network show the efficacy of identifying high-priority sites. Our experiments show that partial assessment results and coarse information reduce the preparedness accuracy, which motivates our multi-dimensional scoring for planning. Future work will calibrate BlackOps with ground-truth data for improved assessments, with a focus on real-time hazards, e.g., fire. We envision that practitioners can use BlackOps to improve healthcare resilience by pre-staging power resources prior to multi-day power outage events.

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